A Reflective Journey: Navigating Your Cumulative Experience at Iowa State University

Collin Van Weelden

I have worked on many projects throughout my time at ISU—some through clubs and some through classes. Working in larger groups, I have been able to work on more complex projects and see the big picture while still being able to design at a low level. The opportunities I was a part of included undergraduate research, co-ops/internships, coursework projects, and lastly my senior design project. My experiences have walked me through the full engineering process enough times that I am able to successfully predict what is expected of me and what the outcomes should look like. This is a perspective that a young engineer needs but that would be completely absent without my education and time at ISU.

I started at Iowa State University in August of 2022 as an aerospace engineering student before switching to electrical engineering in my second semester. In my first semester, I joined an M2I team that was full of very smart and dedicated students, all wanting a similar goal of an impressive project. The team was working with a drone, and this aligned with some of my interests from before college, so I joined. This was one of the most important decisions I made since it showed me what hard-working and dedicated students could really do. The things I was learning in my classes were immediately put to use, and I was forced to research and learn new things to achieve the goals we wanted. At the end of the semester, I was given the most improved award by the team. In the same semester, I joined as an undergraduate research assistant. I worked with senior aerospace engineers and learned some of the basics of controls, CAD, and circuit design. The deepest place of learning for me in this research was how to do underwater communication wirelessly. Through this one problem, I was exposed to many disciplines, from optics to antennas, submarine controls, and history. From these early

experiences, I learned that the best way to grow is by jumping into a project and exposing myself to the engineering behind it. On top of this, it was working on these circuits and controls projects that I discovered I actually wanted to pursue electrical engineering.

A key feature of any engineer is practical skills inside and outside of the classroom. Each experience allowed me to grow in different ways and pick up more of these necessary skills. Through my time working in the ETG, I improved my PCB layout ability, soldering skills, and circuit troubleshooting. When I was a co-op with International Paper, I learned how to manage my time and keep organized. When I was a co-op with Collins Aerospace, I learned PCB design and how to communicate with different teams to get needed paperwork done. Each year, these skills compounded and built upon each other, allowing me to go even further with my learning and application of said learning.

Engineering requires taking initiative and going beyond what is required of you. During my time at Iowa State University, I had the opportunity to practice this skill. An example is the design project in EE 2300 when working on an IR remote. In our design project, my lab mate and I decided to use a different circuit than what was detailed in the lab document and different from what we had used in the labs leading up to this one. We asked the professor if we could work it into our design, and he encouraged it. From this choice, we discovered how the added complexity would change the design in both positive and negative ways. The project ended up being a success, and we learned how to use a 555 timer circuit as a result. Another example is how a classmate and I both learned PCB design during our internship experiences, and we wanted to return that knowledge to the students at ISU. We then worked with IEEE to create a PCB basics and design seminar using free industry tools. This successfully handed down our knowledge to over 40 students over the course of 2 semesters.

In senior design, we are designing a drone from scratch. The end goal is to have made all of the PCBs on the drone as well as the firmware and software needed to support it. We took

a risk by doing a large and challenging project, which forced us to put in a lot of time, and I had to learn skills I didn't already have, like microcontroller layout, how to support a firmware team as a hardware designer, and how to manage burnout. This senior design project has really tested my ability to communicate since I have to communicate with 6 other students, the faculty advisor, and the instructor. Also, we have to be able to present and communicate clearly to the industry panel. Ethics has been a big emphasis in our senior design project, and we have set up a meeting to find out our options for protecting our work from being used in weaponry. This experience has taught me that an engineer is responsible not only for the product but also for how and what that product is used for.

An example of where my skills came together was during the first semester of my senior design project. My knowledge of basic PCB design mixed with knowledge of how a drone works allowed me to design an ESC for the motors. The knowledge made itself apparent when I needed to meet design requirements such as high current ratings and EMI suppression. Knowing about copper thickness options from board manufacturers and also via stitching, along with crosstalk-prevention trace widths, allowed me to complete my design with success. The knowledge of drone frames and motor connections gave me the idea to split the board into two symmetrical ones to simplify layout and assembly.

As time went on, I learned how to work through and solve complex problems. The techniques that I picked up allowed me to visualize and fully conceptualize some of the detailed math and science that goes into topics like signals and systems. When I am stuck on a problem or topic and can't seem to wrap my head around it, I will turn to classmates and professors to see if they can explain it better. YouTube videos or web tutorials can also work. If this doesn't work, then a piece of paper or a whiteboard can allow me to get the ideas in my head out. In general, the more time I spend thinking about the topic, the closer I will get to understanding it. I believe that this method works best for me as it allows me to discover the connections between

my knowledge and the topic I am trying to learn more about since I have to chip away at my gaps in knowledge until it makes sense to me.

If I were to start again, there is not much I would change. Each of the little things that might seem like a waste of time actually enhanced me as an engineer. Things like M2I, where I worked on a team in the aerospace department and just implemented simple control algorithms, later came in handy for senior design when we planned out our drone design. In my undergraduate research in the aerospace department, I did some work with CAD, which later came in useful for senior design again when we wanted to design around what size of drone frame to work with. I am a more well-rounded engineer because of it. However, I do wish that I had discovered my interest in PCB design and also RF/antenna/wave propagation so that I could have taken more high-level classes related to them. If I had another year in undergrad, I would take antenna theory courses and PCB design courses. I would also take high-speed design courses.

I plan to take on projects and become responsible for said projects. I see myself learning and employing more interpersonal skills as I take on more and more complex projects, which require larger teams and stronger communication skills. I will one day become someone's mentor, and I will need to be articulate enough to distill my knowledge to this person. Each of the little skills and disciplines I have picked up over the years will mesh together to create a well-rounded understanding of engineering.